ECE 2305 – Introduction to C Programming
Tutorial 05
Integer, Modulo and Floating Point Division

Write a C++ program to display the results of the following operations.

```
7 / 2
7 % 2
7.0 / 2.0
```

Describe why each of the operations give different results.
Attempt the following operation and describe why it will not work.

```
7.0 % 2.0
```

Input

```
// ECE 2305 Tutorial 05 - Division and Modulo Division
#include<iostream>
using namespace std;
int main()
{       cout << "Division and Modulo Division" << endl;
        cout << endl;

        cout << "7 / 2 ---> " << 7 / 2 << endl;
        cout << "Integer Division returns the whole number part." << endl;
        cout << endl;

        cout << "7 % 2 ---> " << 7 % 2 << endl;
        cout << "Modulo Division returns the remainder parts." << endl;
        cout << endl;

        cout << "7.0 / 2.0 ---> " << 7.0 / 2.0 << endl;
        cout << "Floating Point division returns the complete result" << endl;
        cout << endl;

        //cout << "7.0 % 2.0 ---> " << 7.0 % 2.0 << endl;
        cout << "7.0 % 2.0 ---> error" << endl;
        cout << "Modulo division cannot be done with floating point numbers."
<< endl;
        cout << endl;

        system("pause");
        return 0;
}
```

Output

```
C:\Users\anmal\Desktop\ECE
Division and Modulo Division

7 / 2 ---> 3
Integer Disivion returns the whole number part.

7 % 2 ---> 1
Modulo Division returns the remainder parts.

7.0 / 2.0 ---> 3.5
Floating Point divison returns the complete result

7.0 % 2.0 ---> error
Modulo division cannot be done with floating point numbers.

Press any key to continue . . .
```